Eastern Data, Western Compute in China: Causal Effects on Emissions & Load $\rightarrow$ AI Compute

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Agenda

1. Motivation
2. Original idea & why it isn't feasible
3. Reframed research questions
4. Data needed
5. Identification strategy & model
6. Compute translation (CPC index)
7. Main steps / workflow
8. Risks & open problems

Motivation

Personal interest

- Most AI–economy papers study *downstream* outcomes (innovation, labor, etc.). I focus on *upstream* effects: AI adoption ⇒ more datacenter **compute** ⇒ higher **power demand** and shifting **emissions**.

Why now?

- **Natural experiment (EDWC).** *Eastern Data, Western Compute* shifts workloads west; **staggered commissioning** across provinces gives credible event timing (EDWC).
- **Macro context.** Record renewable build-out while utilisation/curtailment issues persist, plus a public push to expand **compute power** (curtailment/utilisation; record installations; compute power push).
- **Heterogeneity to exploit. CAICT**'s provincial *compute* index enables meaningful cross-province comparisons (CAICT).

Initial idea (dropped): welfare from EDWC-driven curtailment reduction

What I originally proposed

- Estimate **social welfare** from **reducing renewable curtailment** due to EDWC.
- Mechanism: datacenters are **price-sensitive** to ToU/spot $\Rightarrow$ shift flexible workloads to **off-peak** hours where curtailment occurs $\Rightarrow$ less curtailed energy.
- Design: compare **EDWC vs non-EDWC** provinces with an **event study (Sun–Abraham)** around EDWC **cluster go-live** dates.

Why this is <u>not</u> feasible now

- **Binding constraint: no intraday provincial load/price curves.** Without hourly data I cannot show datacenter **price sensitivity** or workload shifting; thus I cannot credibly attribute **curtailment reduction** to datacenters.
- Therefore, the curtailment-based welfare RQ is not defensible with current data.

RQ1 (Simple Alternative): EDWC go-live → provincial emissions & electricity

Research Question: What is the effect of EDWC **go-live** on provincial **CO_2** (daily) and **electricity use** (monthly)?

How I will estimate it (two complementary/possible).

A) Staggered commissioning (Sun–Abraham event-study).

- Use the **Sun–Abraham** cohort-based estimator for **staggered adoption** and heterogeneous effects. Compare **before vs after** go-live for treated provinces *relative to provinces not yet live at the same dates*. **Outcome**: **daily CO_2** and **monthly kWh**.

B) Synthetic Gansu (single-treated).

- Focus on **Gansu** because it has the **cleanest go-live** and the **best data coverage**. Build a **synthetic** Gansu from **non-EDWC** donor provinces in the same **CAICT tier (Q1)**.
- Fit on the **pre-go-live** period; evaluate from go-live onward.
- Report the **post–pre gap** as the causal effect under standard SCM assumptions. **Outcome**: Same as A.

RQ2: How much compute do EDWC-induced changes represent?

Research Question: Given the causal EDWC-induced changes in **CO_2** and **electricity** estimated in RQ1, what is the implied amount of **AI compute** (in exaflop-hours, EF·h) by province and month? Use a provincial **Carbon-per-Compute (CPC)** index to translate energy/emissions into compute units.

What CPC is.

- CPC is an **intensive** metric: tonnes of CO_2 per unit of AI compute.
- We use CPC as a **bridge** to convert RQ1's $\Delta CO_2/\Delta kWh$ into **implied** ΔEF·h.

How we use it.

- Align RQ1's $\Delta CO_2/\Delta kWh$ with CPC inputs to obtain **implied** ΔEF·h and to report **CPC levels/changes**.
- Compare **pre vs. post** within province and vs. controls; assess heterogeneity by EDWC cohort and CAICT tier.

Applied outputs. Implied compute (**ΔEF·h**) by province-month; **CPC index** and changes.

Data needed — RQ1 (outcomes, treatment & controls)

Outcomes

- **Daily CO_2** by province (2019–2025): Carbon Monitor.
- **Monthly electricity consumption (kWh)** by province: *Problem* — no structured dataset; figures are scattered across provincial posts/bulletins. *Solution* — scrape and standardize into a province–month panel.

Treatment & cohorts

- **EDWC timing:** national EDWC announcement date and **province-specific** data-center *go-live* dates to anchor T_0.

Controls

- **Weather** (temperature, precipitation), **statutory holidays** and **industry cycles**.

Additional series to develop

- **Monthly curtailment by province** (e.g., renewable utilization %).
- **Monthly investment by province**, **Monthly employment by province**, etc.

Data needed — RQ2 (CPC translation inputs)

From RQ1 (to translate)

- **Monthly, province-level** ΔCO_2 and ΔkWh around EDWC T_0.

CPC building blocks (exogenous to outcomes)

- **Provincial carbon intensity (CI) by month:** compute as $CI_{p,m} = CO_{2,p,m}/MWh_{p,m}$ (tCO_2/MWh) using the same monthly CO_2 and electricity panels.
- **PUE paths:** province-month trajectories converging to the policy target <1.5 by 2025 (uncertainty ± 0.1).
- **Compute efficiency (GF/W), monthly (AI-equivalent):** Green500 FP64 frontier (interpolated) bridged by HPL-MxP speedups; add MFU realism and a China interconnect haircut (A800/H800 epochs).

CAICT (cross-sectional)

- Province-level **compute index** (adimensional, cross-section) to classify provinces into **Elite/Q4/Q3/Q2/Q1** for *heterogeneity* in CPC and for *donor-pool* construction. CAICT white paper.

Monthly electricity consumption — what & how

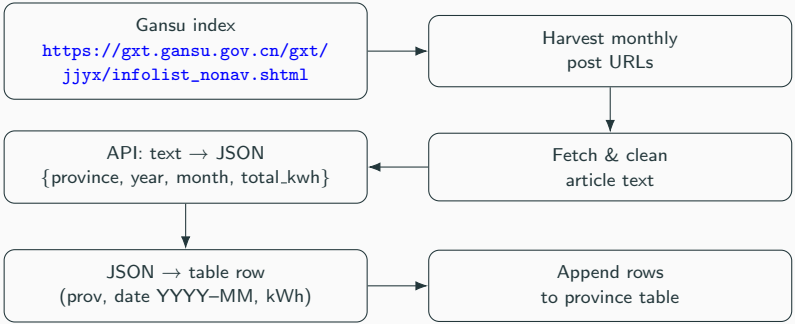
Goal. Province–month panel (2019–2025) of *total electricity consumption* for: (i) EDWC event-time effects (RQ1); (ii) monthly carbon intensity used in RQ2.

Where it lives. Official provincial bulletins (industry/statistics). Example index: Gansu. **Reality.** *No structured dataset.* Values are scattered across monthly posts; formats vary.

Pipeline (reproducible).

- **Source discovery (per province):** find the official list/search page and a recurring title pattern (e.g., "Electricity Production and Operation Status").
- **Link harvesting:** paginate; keep {title, URL, published date} for **monthly** posts only.
- **Text extraction:** download HTML/PDF and clean the article body.
- **API structuring → JSON:** send the cleaned text to a generative API (e.g., *DeepSeek*) with a strict prompt to return only {province, year, month, this-month total kWh (and industrial, if present), YoY, units}.
- **Panel assembly:** merge all monthly JSONs for province p into a tidy *province–month* table; repeat across all provinces p to build the national panel.

Gansu example: bulletin → JSON → table



Example table (Gansu, "this-month" value only).

Province	Date (YYYY-MM)	kWh (this month)
Gansu	2025-08	15,066,000,000

EDWC commissioning timeline (T_0 anchors)

Province	T_0 (commissioning)	Evidence (link)
Gansu (Qingyang)	2024-06	Commissioned; racks in operation (Gansu Gov; Xinhua)
Ningxia (Zhongwei)	2023-07-21	China Telecom Ningxia DC Phase I accepted; ready for operation (Zhongwei Gov; Zhongwei DRC progress)
Inner Mongolia (Horinger)	2024-04-28	China Mobile Hohhot Intelligent Computing Center commissioned/in production (China Mobile; Inner Mongolia News)
Guizhou (Gui'an/Guiyang)	2023-12 / 2024-01	NetEase DC trial run end-2023; ramp in 2024 (municipal/admin) (Guiyang Daily PDF; Gui'an Admin)

Notes: We use *first commissioning* as T_0 (not construction); dynamic windows capture ramp-up.

Estimating monthly ΔCO_2 and ΔkWh (Sun–Abraham)

What we estimate.

- Outcomes (monthly panel 2019–2025): **CO_2** (primary), **kWh** (secondary where available).
- Treatment: province-specific EDWC **go-live** month E_p (T_0); event time $\ell = t - E_p$.

Identification (intuitively).

- At each event month ℓ, compare **treated** provinces to those **not yet treated** at the same calendar month.
- Absorb time-invariant differences with **province FE** (α_p) and national seasonality/shocks with **month FE** (δ_t).
- Minimal controls $X_{p,t}$: **weather** (HDD/CDD, precipitation) and **holidays**.

Why Sun–Abraham (SA) instead of TWFE).

- Handles **staggered adoption** and **heterogeneous** effects across cohorts.
- Avoids TWFE **negative-weight** bias; delivers **event-time** effects β_ℓ that we can plot/average.
- Transparent use of **never-/later-treated** as controls at each ℓ.

Model specification (monthly event study)

Estimating equation

$$y_{p,t} = \alpha_p + \delta_t + \sum_{\ell \in \mathcal{L},\, \ell \neq -1} \beta_\ell \mathbb{1}\{t - E_p = \ell\} \cdot \mathbb{1}\{p \in \mathcal{G}\} + X'_{p,t}\gamma + \varepsilon_{p,t}.$$

Variables (concise).

- $y_{p,t}$: monthly outcome $\in \{CO_2\ (\mathrm{t}),\ \mathrm{kWh}\}$, estimated in **separate** runs.
- E_p: first EDWC **go-live**(T_0) in province p; $\ell = t - E_p$ is **event time**; omit $\ell = -1$.
- $\mathbb{1}\{p \in \mathcal{G}\}$: treated set (Gansu, Inner Mongolia, Ningxia, Guizhou); controls are never-/later-treated.
- α_p: province fixed effects; δ_t: calendar-month fixed effects.
- $X_{p,t}$: weather (HDD/CDD, precipitation) and statutory holidays.
- β_ℓ: **event-time** ATT at lead/lag ℓ (Sun–Abraham, interaction-weighted GTATE).

Estimation details. Window $\ell \in [-12, +18]$ months; tails binned; SEs clustered by province.
Output used later. $\widehat{\beta}_\ell \Rightarrow$ monthly ΔCO_2 and ΔkWh series aligned to each province's T_0.

CPC (Carbon-per-Compute): definition & inputs

Goal. Monthly **CPC** by province: tonnes CO_2 per EF·h of AI compute.

Definition.

$$\mathrm{CPC}_{p,t} \;=\; \mathrm{CI}_{p,t} \;\times\; \left(\frac{1000}{\mathrm{GF/W}_{AI\,\mathrm{China}}(t)}\right) \;\times\; \mathrm{PUE}_{p,t} \quad \left[\frac{\mathrm{tCO_2}}{\mathrm{EF}\cdot\mathrm{h}}\right]$$

Inputs.

- $\mathrm{CI}_{p,t}$: provincial carbon intensity computed from our **monthly CO_2** and **monthly electricity** panels, $\mathrm{CI}_{p,t} = \mathrm{CO}_{2,p,t}/\mathrm{MWh}_{p,t}$.
- $\mathrm{PUE}_{p,t}$: province–month paths converging to China's policy target PUE < 1.5 by 2025 (State Council news).
- $\mathrm{GF/W}_{AI\,\mathrm{China}}(t)$: national monthly AI-equivalent efficiency (GFLOPS/W) we construct once (see next slide) and apply uniformly unless province-specific data appear.

Building $\mathrm{GF/W}_{\mathrm{AI,China}}(t)$: FP64 frontier → AI workload → MFU → China interconnect

Monthly series (2019–2025), auditable:

1. **FP64 efficiency frontier.** For each Green500 edition, take the *Q4* GF/W Green500 (June 2025).
2. **Adapting to AI workload.** Multiply by the *HPL-MxP (HPL-AI) throughput speedup* HPL-MxP results. Assumption (speedup ⇒ efficiency) NVIDIA tensor-core note.
3. **Utilization realism (MFU).** Apply $\kappa_{\mathrm{global}} \in [0.4, 0.6]$ (center 0.5) for practical cluster utilization (context: Megatron-LM).
4. **China interconnect haircut.** $\lambda \in [0.3, 0.5]$ (center 0.4). Define $\beta(t)$ as the *NVLink per-GPU bandwidth ratio* China vs global: A800 400 $\mathrm{GB/s}$ vs A100 600 $\mathrm{GB/s} \Rightarrow \beta = 400/600 = 0.667$ (A100 datasheet). Apply $\kappa_{\mathrm{China}}(t) = 1 - \lambda\,[1 - \beta(t)]$.

Assemble.

$$\mathrm{GF/W}_{\mathrm{AI,global}}(t) = \overline{\mathrm{GF/W}}_{\mathrm{FP64}}(t) \cdot s_{\mathrm{mxp}}(t) \cdot \kappa_{\mathrm{global}}, \qquad \mathrm{GF/W}_{\mathrm{AI,China}}(t) = \mathrm{GF/W}_{\mathrm{AI,global}}(t) \cdot \kappa_{\mathrm{China}}(t).$$

Add uncertainty bands via simple Monte Carlo over $s_{\mathrm{mxp}},\ \kappa_{\mathrm{global}},\ \lambda,\ \mathrm{PUE}$.

Main steps (end-to-end)

1. **Literature review**: EDWC/data centers & power; modern staggered-DiD; energycompute accounting; clarify this paper's niche.
2. **Build province–month kWh panel**: find official monthly bulletins; scrape article pages; structure with a generative API (e.g., DeepSeek) to JSON {`province, year, month, total_kwh`}; validate and export a tidy panel.
3. **Assemble RQ1 outcomes**: aggregate daily Carbon Monitor CO_2 to monthly; join with the kWh panel; harmonize IDs/units; QA and versioning for reproducibility.
4. **Estimate RQ1 (Sun–Abraham)**: monthly event study around datacenter go-live; cohort-interaction weights; event-time plots and post-period averages with province-clustered SEs.
5. **Synthetic Gansu (optional)**: build a CAICT-matched donor pool; fit on pre-period; placebo-in-space distribution; robustness (leave-one-out, alt. windows).
6. **AI efficiency index (GF/W)**: monthly series from Green500 frontier bridged with HPL-MxP speedups; add MFU/utilization and interconnect adjustments; document assumptions/ranges.
7. **CPC construction**: compute provincial monthly carbon intensity (CI) from our CO_2 and MWh panels; choose PUE paths; combine CI, GF/W and PUE to obtain a Carbon-per-Compute index.
8. **Translate to compute & outputs**: convert RQ1 deltas into implied EF·h; publish province-month series, CPC dashboards, and stress-test scenarios.

Problems (risks & mitigations)

1) Monthly kWh coverage.

- *Risk*: Insufficient continuous 2022–2025 series; mixed formats (month vs YTD).
- *What I know*: **Gansu** 100% in 2024–2025; other target provinces look complete for **2025**; final coverage unknown until full scraper runs.
- *Mitigation*: Use CO_2 as primary outcome; acceptance rule $\geq$80% months; strict unit checks.

2) Program fit (Master's in Chinese Economy).

- *Risk*: Perceived as tech/energy rather than economics.
- *Mitigation*: Add **economic outcomes**: provincial investment and employment. Map **investment** $\rightarrow$ **compute** via RQ2 (EF·h priced by GPU-year).